FLEET TEAM DIGEST

737MAX-FTD-10-17001

Issue Title : Wheel Chock Usage

Airplane Model	ATA	Minor Model(s)
737MAX	1010-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
727, 777, 747, 757, 787, 707, 737, 737NG	9E991

Originated Date	Last Revised Date	Created On
05/02/2017	05/02/2017	04/21/2017

Estimated Completion Date	Next Update Date

Applicability

All 737MAX airplanes.

Description

Operators sporadically have questions about wheel chocks and recommended practices concerning wheel chocks. In this article, we will attempt to respond to some of the common questions.

Correct wheel chock usage is important since there have been several incidents and accidents associated with improper wheel chock usage. These events include airplanes rolling into jet stands or adjacent equipment and uncontrolled airplane movement during engine runs.

Background

BACKGROUND

In this article, Boeing would like to reiterate the importance of following the instructions in the AMM's concerning wheel chock usage during normal parking (AMM chapter 10-11-XX) and during engine runs (AMM chapter 71-00-XX).

NORMAL PARKING SITUATIONS

A common cause of unintended airplane movement during normal parking is over-reliance on the parking brake to hold the airplane stationary. For normal parking situations, Boeing does not intend for the parking brake to be used to hold an airplane stationary for extended periods of time since parking brake hold capability will decrease over time as the hydraulic pressure in the brake system bleeds away. Also, the parking brake latching mechanisms can sometimes be worn or not fully engaged, thereby allowing the parking brake to unexpectedly release if jarred or bumped. As a result, an airplane should never be left unattended without chocks installed.

ENGINE RUNS

During engine runs, it is imperative to not rely on wheel chocks. Instead, the parking brake, in conjunction with specified weight/CG requirements, should be used to prevent unintended airplane motion.

Status

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Closed.

Interim Action

N/A

Final Action

N/A

Operator Action

The following provides some Boeing general recommendations concerning wheel chock usage:

NUMBER OF CHOCKS:

The AMM chocking requirements vary slightly from model-to-model, but as a general rule Boeing suggests that a minimum of 4 chocks (2 sets) be used for all airplane models, regardless of how many wheels the airplane has.

CHOCK LOCATION FOR NORMAL PARKING:

The preferred approach is to place a set of chocks on the left main landing gear and the second set of chocks on the right main landing gear. Which tire from each landing gear is up to the operator. On airplanes with main gear trucks with 4 or more tires on each truck, Boeing recommends that the forward chocks be placed in front of the forward tires on the truck and the aft chocks be placed behind the aft tires on the truck (in other words, do not place chocks between the forward and aft wheels).

Some operators prefer to place one set of chocks on one main gear and the second set chocks on the nose gear. This practice is not preferred by Boeing but is acceptable for routine parking situations.

For normal parking, chocks should be placed close to the tires, but not touching the tires (two inches away from the tire is a typical recommendation). This helps prevent the chocks from becoming pinched under the tire if the airplane moves slightly or if the airplane wheelbase changes slightly as the airplane is loaded/unloaded.

CHOCKING FOR ENGINE RUNS:

The coefficient of friction between an airplane tire and the runway surface is generally better than between the wheel chock and the runway surface. If the chock is placed snugly against a tire and the airplane moves slightly forward during the engine run, the tire will tend to roll up onto the chock thereby transferring the frictional forces from the tire and the ground to the chock and the ground (thus decreasing the airplane hold capability). To prevent this, Boeing recommends that the chocks be placed a few inches in front of the front tires when performing engine runs. About 6 - 12 inches is recommended for engine runs. This ensures that the parking brake is the primary device preventing airplane motion, not the wheel chock.

When performing engine runs, it is very important to follow AMM guidelines for airplane weight/CG and runway surface condition since the airplane can slide across the runway surface very easily if the airplane is too light or if the runway surface is slippery.

Part Information

The following provides some Boeing recommendations concerning wheel chock selection:

CHOCK MATERIAL:

There are four common wheel chock materials: Metal, Rubber, Urethane, and Wood. Each of these materials has advantages and

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disadvantages. We have prepared the attached table that summarizes the Boeing opinion of the merits of each material. SAE document AIR4905 contains very similar information.

CHOCK HEIGHT:

The height of the chock is primarily a function of the size of tire to be chocked. For the 737, a typical main gear chock height would be about 6 or 7 inches. For airplanes with large tires (greater than 50 inches or so) it would be preferable to use taller chocks.

WHEEL CHOCK LENGTH:

Some wheel chocks are long and can secure the two tires on one axle at the same time while others are short and secure only one tire. Boeing considers either long chocks or short chocks acceptable, but some operators have indicated that the long chocks are excessively heavy and cumbersome for maintenance personnel. Whether long chocks that straddle two tires or short chocks that only straddle one tire are used, the above noted usage recommendations are the same.

CHOCK ANGLE:

Chocks should have an angled surface where they contact the tire and this angle should be about 45 degrees for optimum function.

CHOCK COLOR:

Brightly colored chocks have the advantage of being more noticeable and therefore are less likely to be inadvertently left installed or left on the ramp.

CHOCK MANUFACTURERS:

There are many sources for aircraft wheel chocks and each AMM contains one or more specific recommendations.

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability

Attachments:

a) Wheel Chock Material Comparison

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